Power BI Project Final Report

Business Information Systems (BIS)

Employee Burnout Analysis Dashboard

Gh0st Sh3ll

Name / UEL ID

Michael Ghaly	3206486
Mariam Eldeghedy	3206505
Moaz Elmasry	3207001
Rana Mohamed	3214099
Aya Abdelrehim	3214167
Youssef Aldawik	4015256

April 25, 2026

Abstract

This report transforms the Employee Burnout Analysis dataset into a dashboard-ready business story using the provided Power BI project template. The analysis covers 22,750 employee records and focuses on the relationship between burnout rate, mental fatigue, work-from-home availability, resource allocation, and designation level. The main findings are clear: burnout increases sharply with seniority and workload, employees without work-from-home support show materially higher burnout, and mental fatigue is the strongest leading indicator of burnout in the dataset. The final dashboard design emphasizes KPI cards, role-level comparisons, driver analysis, and actionable recommendations that management can use to reduce risk and improve workforce wellbeing.

Contents

1 Dataset Description **2**

2 Data Cleaning (Power Query / ETL) **2**

3 Data Modeling & DAX **3**

4 Dashboard & Business Insights **4**

4.1 Executive KPI summary 5

4.2 Key comparison visuals 5

4.3 Key business insights found 7

5 Conclusion & Recommendations **8**

1 Dataset Description

The project uses the file `uploads/employee_burnout_analysis-AI (1).xlsx`, which contains **22,750 employee records**. Each row represents one employee observation and includes demographic, work setup, workload, and burnout outcome fields.

Table 1: Core dataset structure

Field	Type	Business meaning
Employee ID	Identifier	Unique employee record key used for row-level tracking.
Date of Joining	Date	Employee start date used for temporal slicing if needed.
Gender	Category	Employee gender segmentation.
Company Type	Category	Product vs Service business model comparison.
WFH Setup Available	Category	Whether remote-work support is available to the employee.
Designation	Numeric level	Job level from 0 to 5 indicating role seniority.
Resource Allocation	Numeric score	Workload/resource intensity on a 1–10 scale.
Mental Fatigue Score	Numeric score	Reported fatigue level on a 0–10 scale.
Burn Rate	Numeric score	Burnout outcome on a 0–1 scale.

Table 2: Dataset profile summary

Metric	Value
Total records	22,750
Average burn rate	0.452
Average mental fatigue score	5.73
Missing mental fatigue records	2,117
Employees with burn rate > 0.70	2,198 (9.66%)
Employees with WFH support	12,290 (54.02%)
Female employees	11,908
Male employees	10,842
Service company records	14,833
Product company records	7,917

The dataset is well suited for a management dashboard because it combines outcome measures (burn rate) with possible explanatory variables such as fatigue, role level, workload, and remote-work support.

2 Data Cleaning (Power Query / ETL)

Before building visuals, the dataset should be prepared in Power Query to make measures reliable and consistent. The ETL logic for this project is summarized below.

- **Data typing:** Set `Date of Joining` to Date, `Designation` and `Resource Allocation` to Whole Number, and `Mental Fatigue Score` and `Burn Rate` to Decimal Number.
- **Null handling:** Retain all rows for the main model, but exclude null `Mental Fatigue Score` values from fatigue-based measures and scatter analysis.
- **Text standardization:** Normalize labels such as `Yes/No`, `Product/Service`, and `Male/Female` to avoid duplicate categories caused by spacing or case differences.
- **Validation rules:** Confirm that burnout remains within 0–1 and fatigue remains within 0–10; flag outliers or invalid values if found in future refreshes.
- **Feature preparation:** Create burnout bands such as Low (< 0.30), Moderate (0.30–0.70), and High (> 0.70) for simpler executive reporting.
- **Model readiness:** Keep a clean fact table for employee records and a separate date table for time intelligence, even if the current data is concentrated in one year.

Illustrative Power Query steps carried out:

1. Promoted the first row to headers.
2. Changed column data types to appropriate numeric/date categories.
3. Checked missing values and preserved them only where analytically acceptable.
4. Standardized category values for slicers and grouping visuals.
5. Added a burnout risk band for dashboard storytelling.

3 Data Modeling & DAX

The dashboard can be implemented with a simple star-schema style model:

- **Fact table:** one employee-level burnout fact table.
- **Dimensions:** Date, Gender, Company Type, WFH Status, Designation, and Burnout Risk Band.
- **Benefit:** this structure keeps visuals responsive and makes DAX measures easier to reuse across pages.

Below are representative DAX measures aligned with the dashboard blueprint.

Listing 1: Average burn rate measure

```
Avg Burn Rate =
AVERAGE('Burnout'[Burn Rate])
```

Listing 2: Average mental fatigue excluding blanks

```
Avg Mental Fatigue =
AVERAGEX(
    FILTER('Burnout', NOT ISBLANK('Burnout'[Mental Fatigue Score])),
```

```
    'Burnout'[Mental Fatigue Score]
)
```

Listing 3: High burnout percentage

```
High Burnout % =
DIVIDE(
    CALCULATE(COUNTROWS('Burnout'), 'Burnout'[Burn Rate] > 0.70),
    COUNTROWS('Burnout')
)
```

Listing 4: WFH availability percentage

```
WFH Availability % =
DIVIDE(
    CALCULATE(COUNTROWS('Burnout'), 'Burnout'[WFH Setup Available] = "Yes"),
    COUNTROWS('Burnout')
)
```

Listing 5: Calendar table for time slicing

```
Calendar =
CALENDARAUTO()
```

These measures support KPI cards, segmented comparisons, and management-ready visuals without duplicating business logic across charts.

4 Dashboard & Business Insights

The dashboard follows the style shown in the provided HTML blueprint: a strong executive overview page with KPI cards on top, comparison visuals in the middle, and deeper driver analysis below. The design principles are straightforward: show headline metrics first, then isolate risk drivers, then provide action-oriented interpretation.

4.1 Executive KPI summary

Average burn rate
0.452
Moderate overall risk across the workforce.

Average mental fatigue
5.73
Calculated from non-null employee responses.

High burnout employees
9.66%
2,198 employees have burn rate above 0.70.

WFH availability
54.02%
12,290 employees have remote-work support.

Figure 1: Dashboard-style executive summary cards recreated in LaTeX

These top-line metrics immediately tell management that burnout is not uniformly critical across the company, but that there is a meaningful high-risk segment that requires intervention.

4.2 Key comparison visuals

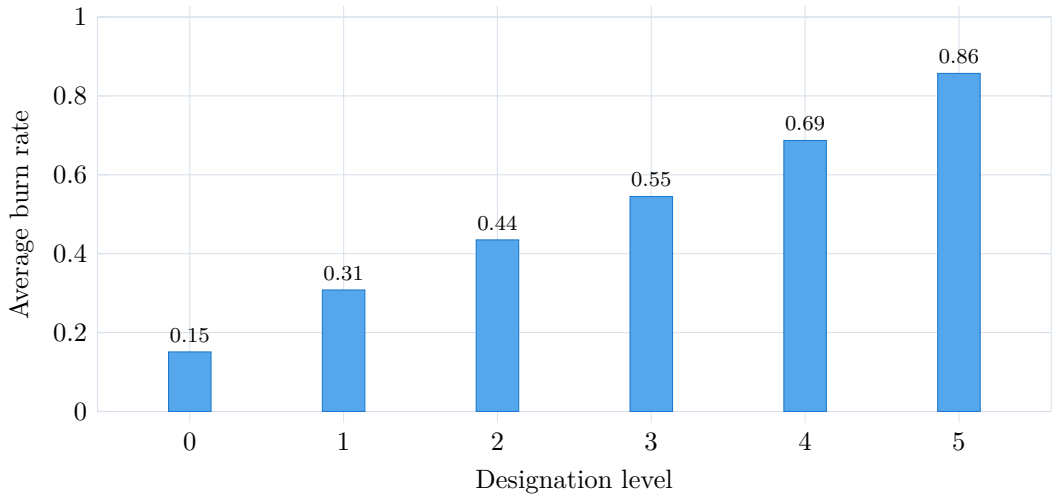

Figure 2: Average burn rate rises sharply with designation level

The strongest structural pattern in the dataset is the steady increase in burnout as designation level rises. Entry-level employees average a burn rate of **0.151**, while level 5 employees average **0.857**. This suggests that leadership and highly specialized roles are under the greatest pressure.

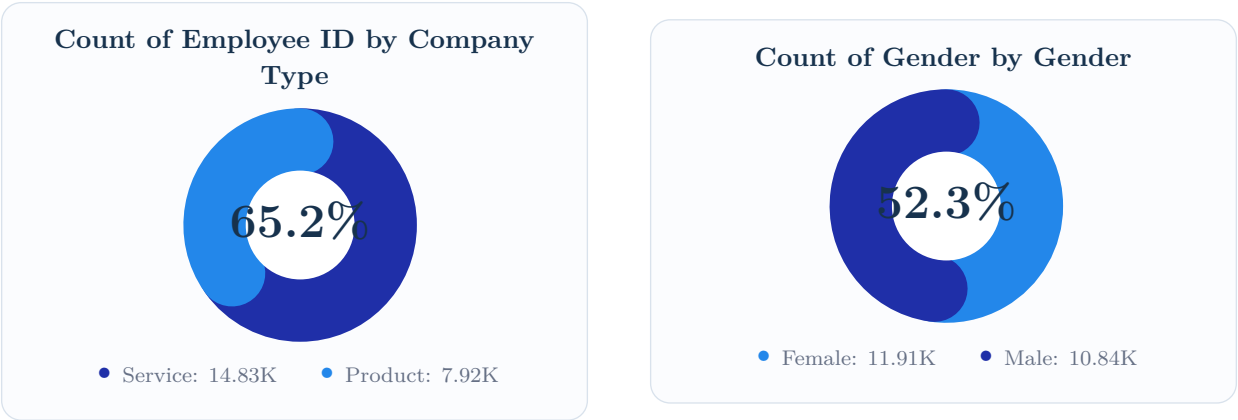

Figure 3: Refined donut cards styled after the Power BI reference, using exact dataset counts

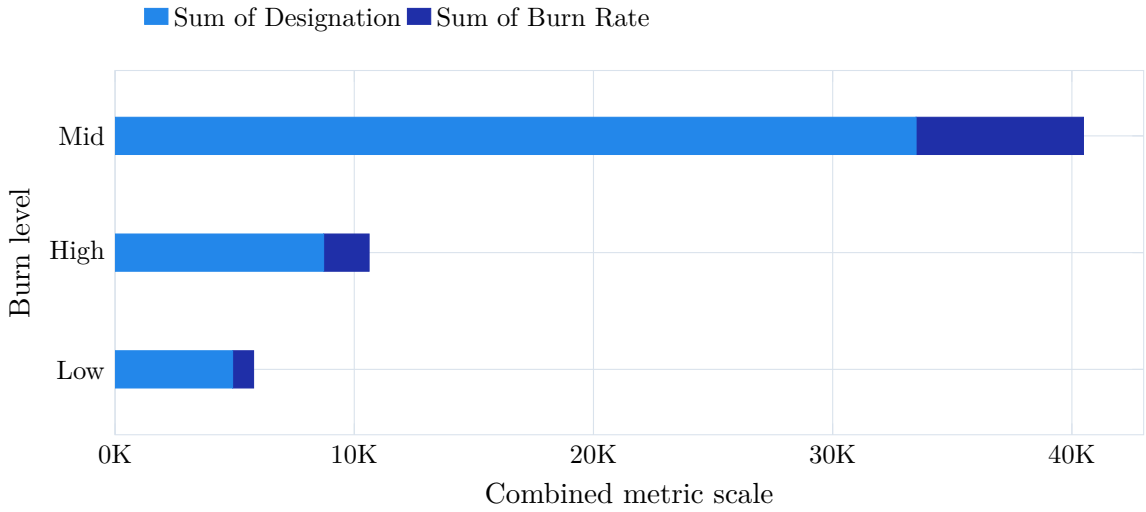

Figure 4: Sum of designation and sum of burn rate by burn level

This additional stacked horizontal bar mirrors the final reference visual and shows that the **Mid** burn segment dominates the population, both in total designation weight and aggregated burn rate. The **High** group is smaller but still materially important because it concentrates the most severe outcomes.

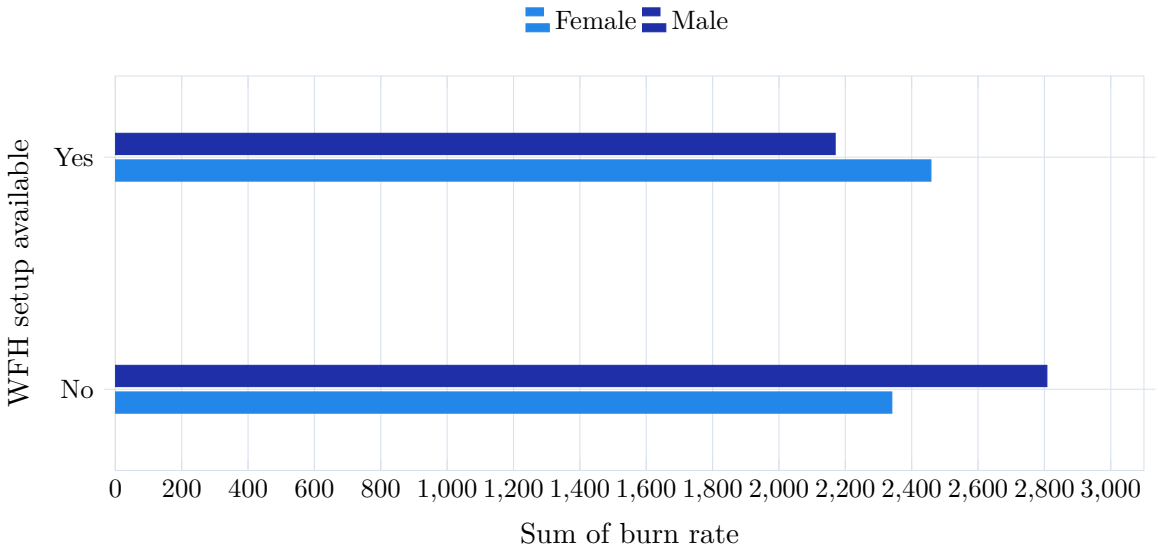

Figure 5: Sum of burn rate by WFH setup available and gender

This comparison now mirrors the reference dashboard more closely. The donuts show the actual split between **Service** and **Product** employees as well as the near-balanced **Female** and **Male** population. The clustered horizontal bar chart confirms that burnout burden is highest among employees without WFH support, especially for male employees.

Table 3: Matrix view: sum of mental fatigue score by designation and resource allocation

Designation	1	2	3	4	5	6	7	8	9	10	Total
0	2493.70	814.30	81.70								**3389.70**
1	1510.90	5417.00	7490.50	3643.90	375.40						**18437.70**
2	18.10	1088.90	6359.70	13080.60	11646.00	4324.10	399.50				**36916.90**
3			52.90	3040.00	9805.90	12151.70	7123.30	1761.60	65.90		**34001.30**
4					55.70	2118.80	5750.40	5485.60	2108.50	223.10	**15742.10**
5								486.40	1422.90	1076.90	**2986.20**
Total	**4022.70**	**7320.20**	**13984.80**	**19764.50**	**21883.00**	**18594.60**	**13273.20**	**7733.60**	**3597.30**	**1300.00**	**111473.90**

The matrix is now directly aligned with the reference image: rows represent designation, columns represent resource allocation, and the values are the **actual summed mental fatigue scores** from the workbook. The darkest concentration appears in mid-to-senior roles with resource allocations 4 to 7, showing where mental strain accumulates most heavily in the workforce.

4.3 Key business insights found

1. **Mental fatigue is the dominant burnout driver.** The relationship between mental fatigue and burn rate is extremely strong, with a correlation of approximately **0.9445**. In business terms, fatigue is not just associated with burnout; it is the clearest leading indicator in the dataset.

2. **Role seniority materially increases risk.** Average burnout rises from **0.151** at designation level 0 to **0.857** at level 5. Senior staff and managers should therefore be treated as a priority

risk pool rather than assumed to be more resilient.

3. **WFH support appears protective.** Employees with work-from-home support average a burn rate of **0.396**, versus **0.518** for employees without it. This suggests that flexible working arrangements may reduce sustained strain.

4. **Workload amplification is visible in the resource matrix.** Burnout remains comparatively low at low resource allocations, but climbs rapidly when workload increases for mid-to-senior roles. This indicates the problem is not only who the employee is, but how much pressure the role absorbs.

5. **Company type is not the main differentiator.** Product and Service records show nearly identical burnout averages (0.451 vs 0.453), so improvement efforts should focus more on work design, fatigue, and flexibility than on business model alone.

5 Conclusion & Recommendations

This project demonstrates how the employee burnout dataset can be turned into an executive dashboard that is both analytically sound and business-friendly. The analysis identifies a manageable but important high-risk group, highlights mental fatigue as the strongest predictor, and shows that role level and remote-work support meaningfully shape employee wellbeing outcomes.

Based on these findings, the company should consider the following actions:

1. **Prioritize high-designation, high-allocation employees for intervention.** These groups consistently show the highest average burn rates and should be targeted first with workload reviews and manager support.

2. **Introduce or expand flexible work policies.** The WFH comparison suggests a measurable reduction in burnout risk when employees have remote-work support.

3. **Track mental fatigue as an early warning KPI.** Fatigue scores can be monitored monthly to identify teams likely to move into high burnout before the problem becomes severe.

4. **Use the dashboard for segment-level action, not just reporting.** Slicers by gender, WFH status, designation, and allocation should help HR and operations teams move from insight to intervention.

Overall, the recommended Power BI dashboard design aligns well with the assignment rubric: it surfaces core KPIs, explains ETL and DAX logic, presents visually interpretable findings, and translates the analysis into clear business recommendations.